

“Simulating the effects of different gravities found on foreign planets to evaluate the potential of growing life elsewhere in space”

Research question: How do different gravities, simulated by a clinostat, affect the germination of seeds measured by root length of bittercress (*Lepidium sativum*)?

Subject: Biology

Word Count: 3146

Session: May 2024

INTRODUCTION	3
RESEARCH QUESTION	5
VARIABLES	5
HYPOTHESIS	6
METHODS & APPARATUS	6
Calculations	7
DATA COLLECTION	8
DATA PRESENTATION AND ANALYSIS	9
Data graphs	9
Data analysis	10
Correlations between root length and g-force	11
CONCLUSION	11
EVALUATION	12
DISCUSSION	13
APPENDIX	15
Rotating trials of the zero-g group and mars group	15
Rotating trials of zero-g group and mars group	15
Rotating trial of control group, fully parallel with the ground	15
Visible condensation on lid of petri dishes	16
BIBLIOGRAPHY	16

INTRODUCTION

The word ‘colony’ has a deep and rich background in the history of the Americas. The word originates from the latin phrase ‘colonia/colonus’ meaning ‘settlement,farm/settler,farmer’. In order to colonize a new land, a permanent establishment needed to be set and one of the defining factors of a permanent residence was growing crops, which took years. Setting our sights on the stars, a space colony would essentially mean a permanent establishment in... well, space. This would put an even higher emphasis on growing crops than the Pilgrims had as we would be in an entirely inhabitable environment. Intense research is needed for us to figure out how to grow crops on exoplanets (planets outside of the solar system). Starting with the variable of a planet that we cannot control: gravity. A clinostat, which is a constantly, slowly rotating probe can be used to simulate zero-gravity by constantly turning the test-subject on its head: not allowing the test subject to distinguish up from down [8].

Going back to growing crops. Before a plant can even undergo photosynthesis and properly grow, it first needs to germinate. Germination is a complex molecular process that allows the plant to grow at the appropriate timing. During germination DNA is repaired, mitochondria are repaired and synthesized, proteins are transcribed from existing mRNA fragments and most importantly a radicle is formed which becomes the shoot and root of the plant.

Gravitropism, the ability of a plant to respond to the stimulus of gravity, allows the plant to grow its stem upwards towards sunlight and its roots downwards towards the mineral rich soil underneath. Gravitropism is arguably one of the most essential abilities of organisms of the plantae kingdom. The key lies in dense auxin hormones in the growing cells of the shoots and roots. These dense granules (see Figure 1) sink through the cytosol of the cell, passing through microfilament structures to the bottom of the cell [10]. Next, through an intricate biochemical process the plant (in the example of roots) grows its root-cap in the direction of the auxin hormone accumulation.

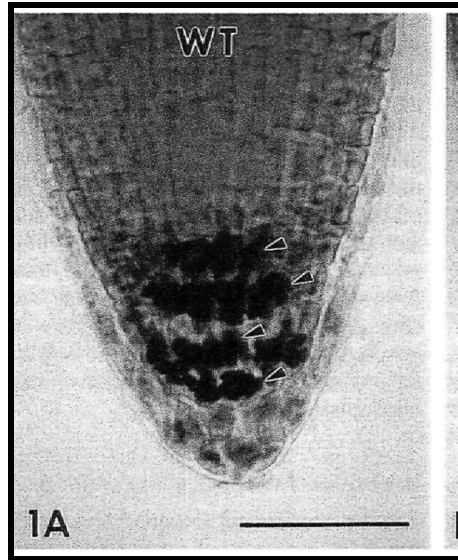


Figure 1. Dense auxin hormone granules located in columella cells of the root cap (Kiss J., 1996).

The problem of growing food that arises with space colonization is how do the different gravities affect the ability of a plant to grow or even shoot in lower or zero-gravity. Can plants even germinate in space where they would not be able to rely on their gravitropic tendencies? To answer this question many experiments in space need to be conducted. However, since the cost of transport of just one kg of material today is: \$20,000 [5], this is an extremely impractical way to answer this question, which would take a lot of time, money and manpower to solve. And what about experimenting how such crops would grow on extraterrestrial planets? Would we first send crews to research if we can even stand a chance of colonizing an exoplanet? No, that would be silly. This gives rise to the need of producing an accurate simulation of sub-g gravity here, on Earth so we do not have to make long and difficult space trips to collect data.

Which brings us back to the clinostat, a nifty device that allows us to simulate zero-g on plants by essentially constantly rotating the dense auxin hormones in the columella of the root-cap (see Figure 2), confusing it and inhibiting it from using gravitropism to grow. To simulate gravities larger than zero-g, yet smaller than Earth's gravity (so called sub-g gravities) one can set the clinostat at an angle perpendicular to the ground. This accumulates a net gravity on the plant calculated by the formula : Angle of rotation

$\pm 0.1^* = \sin^{-1}(G)$, allowing us to simulate a wide range of gravities, as long as they are smaller than that of Earth's.

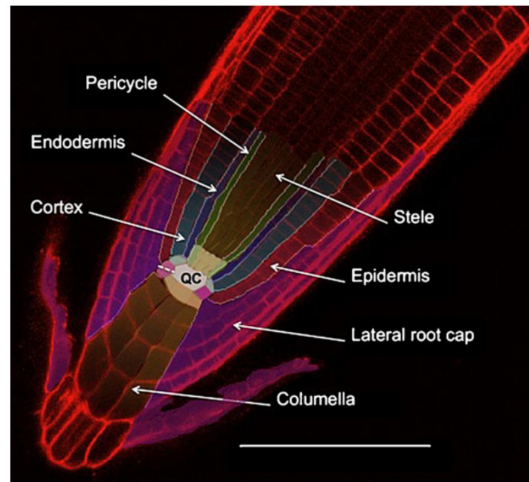


Figure 2. Physiology of the root cap of plants (Arnaud C., 2010).

In light of these facts, this essay will try to answer the question: How do different gravities, simulated by a clinostat, affect the germination of seeds measured by root length of bittercress (*Lepidium sativum*)?

Preliminary Study

Since I did not have access to an industrial clinostat, I had to build it myself. Before summer vacation I ordered all of the necessary parts (motors, cables, bread board, arduino uno and switches). Since I would have to write the program for the device, I started working on that before the parts came. When the parts finally did arrive, I began prototyping the clinostat. Very quickly things started to fall apart. Off the bat it turned out that I ordered the wrong motor. Instead of a continuous motor I got one that could only go 180 degrees in one direction. This forced me to reorder different motors and wait for their delivery. In addition, my code ended up not working at all. It took me another 2 weeks to get the software running smoothly. Lastly, when all was built I was having difficulties keeping the petri dishes attached to

the motors when they were at a steep angle. Nothing seemed to work, and the saying “if it doesn’t stick, you’re using too little duck tape” also failed, as I had no more space to tape anything to.

After a long struggle I was finally able to solve all of the problems of the prototype; I got the correct motors, cleaned up the wiring, put the device in a nice box making it more robust, got the software to work and got the petri dish to stick by using super glue to make them stick and alcohol to later detach them and finally run a test trial. Now primed with the knowledge gained by my preliminary study and finally being in possession of a functional clinostat (see Figure 3), I started planning my full experiment.

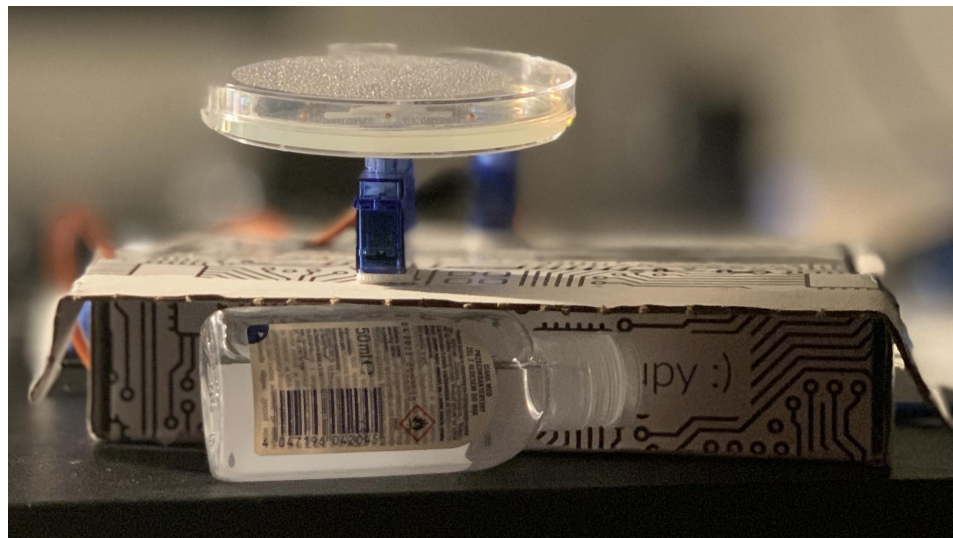


Figure 3. The final version of my clinostat contains three motors (blue boxes) capable of being set at various angles.

RESEARCH QUESTION

How do different gravities, simulated by a clinostat, affect the germination of seeds measured by root length of bittercress (*Lepidium sativum*)?

VARIABLES

Independent variables:

- Exposure to different net forces of gravity of germinating seeds (0.166g, 0.379g, 0.8g, 0g, 1g)

Dependent variables:

- Root length after 3 days of growth measured using ruler (± 1 mm)

Control variables:

- Humidity of air,
- Temperature
- All groups were held in darkness for the duration of the experiment,
- Speed of the clinostat was kept constant for all groups,
- Healthy seeds were used for the experiment, seeds with an “off” color were rejected from the sample.
- Number of repetitions per group – 9
- Seed position in petri dish
- Seed orientation (micropyle)

HYPOTHESIS

Gravity is used as a means of orientating the plant as it sprouts. Through gravitropism, the plant shoots its roots downwards, deep into soil where minerals and water can be found and its stem and leaves upwards, where it can access light needed for photosynthesis. Recent studies in space have shown [16]

that despite weak gravity, plants are still able to shoot roots and stems in opposite directions. However, the tissue of the plant grows much smaller and weaker due to the plant adapting to its surroundings [3]. For these reasons I hypothesize that the lower the net gravity exhibited on germinating seeds, the shorter the roots will be.

METHODS & APPARATUS

The experiment was conducted in amateur lab conditions. I did not have access to a clinostat, so I needed to create my own. I bought an arduino uno kit along with a servo motor, bought the bittercress seeds, got petri dishes from my parents laboratory and used gelatine powder as an agar substitute. Next after acquiring the unordinary materials, I gathered the home supplies I would use to conduct the experiment. I prepared a kitchen scale ($\pm 0.1\text{g}$), a pot, water, a small spoon, a stove, a ruler, a marker, tweezers, double-sided tape and a calculator. I began with creating the clinostat. I used the box that the parts came in as the shell for the device. I taped a breadboard to the base and connected the arduino uno (a basic computer) to the three motors, allowing me to test three variables at once (see Figure 3). Next I wrote a basic program for the computer to rotate the motors at a constant velocity and taped the motors to different sides of the box. By placing two of the motors on the flaps of the box, I could set different angles of rotation for the groups.

Then I began preparing the petri dishes for the experiment by dissolving 15 g of gelatine powder in 500ml of boiled water, pouring the mixture into each petri dish half full and putting it into the fridge for a day to harden. Afterwards I removed the petri dishes and wiped off condensated water off of the lids and began preparing the watercress seeds.

Seeds were chosen for the experiment based on shape, color, size and purity of the outer shell. This was to insure that only healthy seeds were used for the experiment, lowering the chance of an unsuccessful germination which would skew results of the experiment.

I printed out a template for the placement of the seeds (as different placements would alter the effect of the clinostat on the subjects) and planted 9 seeds in each petri dish in a 3*3 square layout. When the seeds were ready, I taped them to the clinostat and angled the motors according to my calculations to get the desired net g-force and turned on the motors.

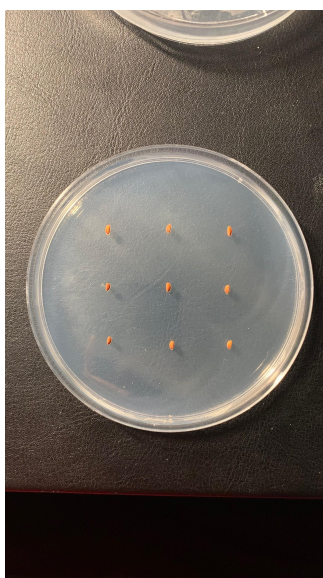


Figure 4. 3x3 square layout of Lepidium sativum seeds on a Petri dish filled with jelly.

During the experiment, I would stop the motors every 12h to take a picture of the plants' progress and I would quickly start them again. After 3 days of rotations, I stopped the motors, took a final photo and discarded the petri dishes.

As safety is a priority, the clinostat was only plugged into the power when everything was set up, proper caution was used while boiling water and the contents of the experiment were properly discarded using alcohol to kill any potentially growing fungi and bacteria as well as making sure to not inhale any particles from the petri dishes.

Calculations

$$\text{Angle of rotation } \pm 0.1^\circ = \sin^{-1}(G)$$

$$\text{Moon} = 1.625 \text{ m/s}^2 = 0.166 \text{ G} \quad \Rightarrow \quad \sin^{-1}(0.166) = 9.6^\circ$$

$$\text{Mars} = 3.721 \text{ m/s}^2 = 0.379 \text{ G} \quad \Rightarrow \quad \sin^{-1}(0.379) \sim 22.3^\circ$$

$$\text{Proxima Centauri b}^* = 8.5 \text{ m/s}^2 = 0.867 \text{ G} \quad \Rightarrow \quad \sin^{-1}(0.867) = 60.1^\circ$$

$$\text{Micro-g} = 0 \text{ m/s}^2 = 0 \text{ G} \quad \Rightarrow \quad \sin^{-1}(0) = 0^\circ$$

$$\text{Earth} = 9.807 \text{ m/s}^2 = 1 \text{ G} \quad \Rightarrow \quad \sin^{-1}(1) = 90^\circ$$

$$\text{Calculation of error} \quad \Rightarrow \quad 1^\circ = \sin(1) = 0.0175 \text{ G}$$

DATA COLLECTION

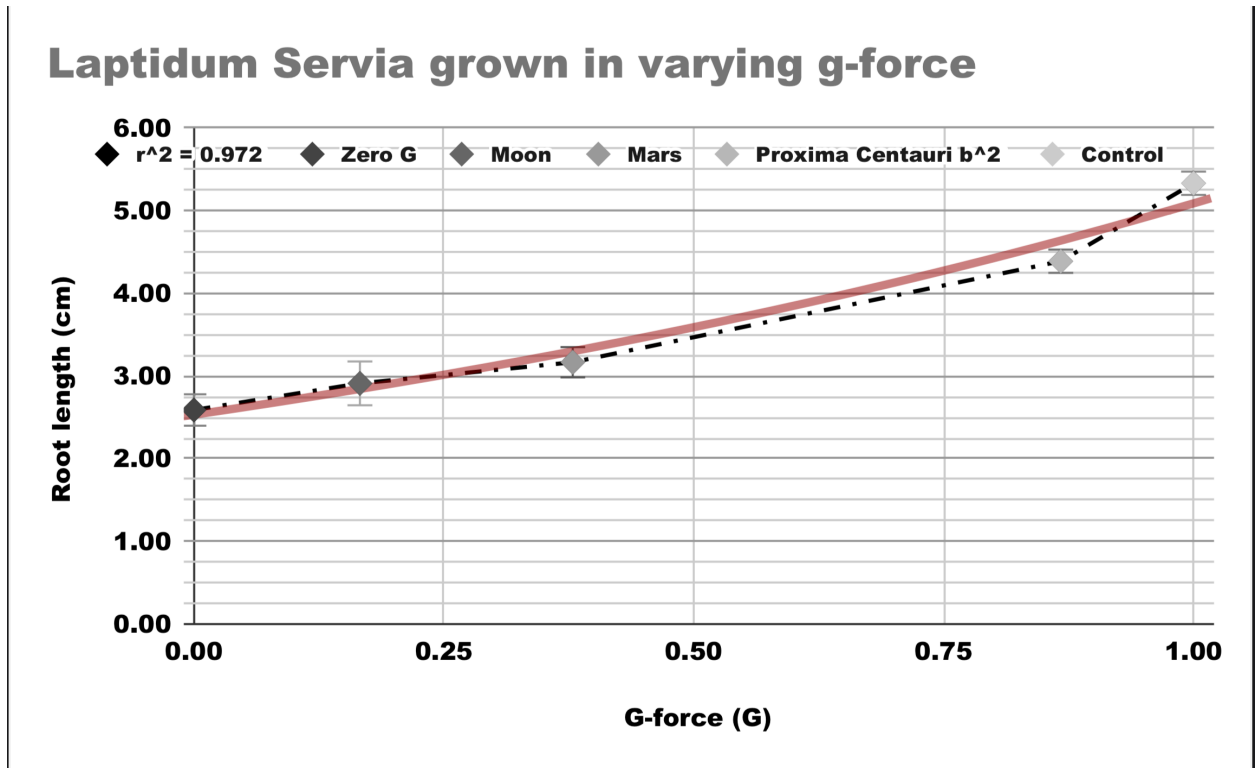
In the table shown below (Table 1) the results of each group are presented along with its average height and the standard deviation. Each row represents a different seed from each group and the columns are arranged with increasing g-force, which can be seen in the second row. All of the cells in the middle of the table show the length of the root in centimeters. The length of the root was measured from the seed to the end of the longest extending radicle.

Seed number		G-force exposure for 3 days ($G \pm 0.0175$)				
		0 (zero-g)	0.166 (moon)	0.379 (mars)	0.867 (proxima centauri b^2)	1 (Control group)
Height ($cm \pm 0.1$)	1	2.5	2.8	3.5	4.3	5.3
	2	3.1	2.7	3.4	4.5	5.2
	3	2.4	3.0	2.9	4.4	5.4
	4	2.8	3.1	3.0	4.7	5.2
	5	2.5	2.6	2.7	4.5	5.3
	6	2.2	2.9	3.1	4.2	5.6
	7	2.5	3.2	3.2	4.1	5.3
	8	2.6	2.9	3.4	4.5	5.2
	9	2.7	3.0	3.3	4.3	5.5
	Average	2.59	2.91	3.17	4.39	5.33
	Standard deviation	0.257	0.190	0.265	0.183	0.141

Table 1. Raw data from the experiment containing 9 tests per group of different exposure of g-force along with the averages for each group and the standard deviation. The results were the root's length measured in centimeters.

DATA PRESENTATION AND ANALYSIS

Data graphs

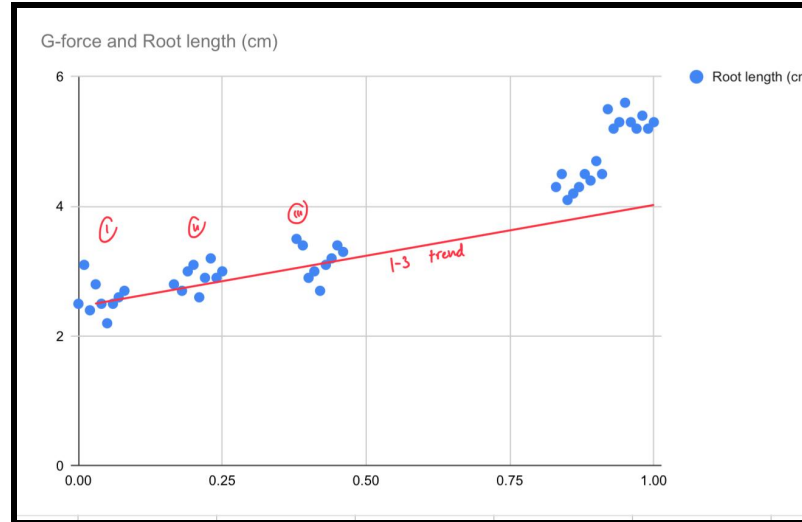


Graph 1. Averages from table 1 illustrated on a graph with a trend line showcasing the trend.

In addition the graph below (Graph 2) shows each individual seed marked on a graph to showcase the positive correlation between test subjects from each group (although a little bit less accurate in order to represent each seed individually).

Data analysis

Graph 2 presents the length of the roots of each individual seedling. The groups are set in order of increasing g-force. The graph also shows a positive trendline, which serves as evidence that with the exposure of higher g-forces the root length of watercress increases. Furthermore, the final two groups can be seen to grow much longer compared to where they would be if the trend of the first three groups was maintained (see Graph 3). This implies that the growth rate of roots of watercress increases exponentially (not linearly) following the increase of g-force.



Graph 3. A trend is drawn on Graph 2 based on the first three groups. An exponential trend line can be clearly observed as a result.

Graph 2. illustrates the averages of root lengths plotted against their exposure to varying g-forces. Standard deviation has been drawn for each group. The graph shows that there is a positive correlation between g-force and the ability of *Lepidium Servia* to germinate and grow roots.

Correlations between root length and g-force

Using the google sheets program the R-squared value for the exponential regression was calculated between the mean of each group for the root length and the g-force applied to each group. This results in a R-squared value of 0.972, which shows a strong correlation between g-force and the length of the roots of watercress seeds.

CONCLUSION

The experiment showed that the growth of roots of *Lepidium sativum* is strongly linked to the force of gravity exhibited on the germinating seed. This aligns with the previous findings of John Z. Kiss, Konathan B. Wright and Timothy Caspar [11] as well as Stanislav Vitha, Ming Yang, Fred D. Sack, John

Z. Kiss [18]. The results serve as evidence that my hypothesis is true since the correlation between the exposure of g-force and the length of the root shoots is positive. This also implies that at higher gravities than that on Earth, plants may have an easier time growing their roots deep into extraterrestrial soil. These results may have an implication with the maturing process of animal embryos, where similar processes (ie. blood circulation) could also be affected by the different forces of gravity. Every seed was able to germinate successfully, showcasing that the selection process of seeds for the experiment was effective in obtaining reliable data to examine relationships between the variables.

EVALUATION

The experiment was conducted in amateur lab conditions, taking place in my basement. However, precise tools were used for measurements and controlling variables which meant that the conceived data was reliable. All calculations were done on my graphic calculator and google spreadsheet eliminating any calculation errors during data processing. In addition, an exponential correlation coefficient was calculated to see whether the two variables (g-force and root length) were correlated to each other, to which the calculation proved that the force of gravity on the germinating seeds had an effect on how long their roots would grow.

As always, there is room for improvement in the methodology of the experiment. Foremost, in future experiments a different clinostat design could be used to further more accurately simulate zero-g conditions. Primarily a slower and more smoothly turning motor could be used to reduce any potential centrifugal forces that may have occurred in the seedlings. Another aspect that could be improved upon would be to increase the investigation groups past that of Earth's gravity. This would require a more complex (3D) clinostat [4], which would need a higher allocation of time and resources but could give us paramount data on a higher variety of potential extraterrestrial environments for crop growth. Lastly, other crops could be investigated ie. potatoes or wheat, which wouldn't be limited to petri dishes if a higher quality clinostat was used.

As for the strong aspects of the methodology they are the high number of trials (9) used for each group, good control of extraneous variables and the reliable clinostat design. These aspects in conjunction were the root cause of the attainment of reliable data that allowed me to come to a solid conclusion. As a final remark the conclusion was supported by numerous experiments and analysis conducted by researchers both on Earth and even in outer space, proving the clinostat to be an accurate low gravity simulator and my results to be accurate and reliable.

DISCUSSION

Growing crops in space would be one of the first challenges tackled by space adventurers in the future. It is for this reason that research on the ability of plants to grow in different conditions than that found on Earth is crucial for humankind to become a multi-planet species in the near future. In the research that I had undertaken I found that there is a positive correlation between the varying g-force experienced by the seeds (simulated by clinostats) and their ability to germinate and grow roots. My findings are supported by other papers like John Z. Kiss et.al. [11]. In their research they used starch mutants of *Arabidopsis* to assess the effect of starch in gravity perception by plants. With higher amounts of starch, the plant's hormones would sink more profoundly to the bottom of the columella cells, which through inducing growth would allow the plant to experience gravitropism. With lower starch levels (meaning a lower density of hormones affecting root growth), less granules could be observed in the columella and less gravitropism could be observed, despite still containing some amounts of necessary granules for invoking growth. This was most likely due to the fact that microfilaments also play a role in the ability of auxin hormones to freely sink in the columella cells, affecting germination and growth according to Guichuan Hou, Deepti R. Mohamalawari, Elison B. Blancaflor [9]. To be more precise the actin filaments form a complex cytoskeleton in the cell. This cytoskeleton is needed for complex cellular processes. However, these protein fibers stand as an obstacle for auxin granules to sink to the bottom of

the cells they are located in. This delay can cause the cell to not be able to grow if for example it is constantly rotated around a single axis.

However, all of these experiments have been done on small plants. This could have led to a bias in the gathered research as there is a possibility that larger plants exhibit more complex cellular processes, possibly undermining the factors that have been researched to this day. Thus it is vital to further research this topic so more precise and general conclusions can be brought forth. Obtaining such data is a necessity, if we wish to become a multi-planetary species.

APPENDIX

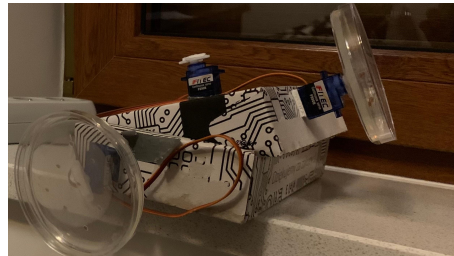


Figure 5. Rotating trials of the zero-g group and mars group.



Figure 6. Rotating trials of zero-g group and mars group.

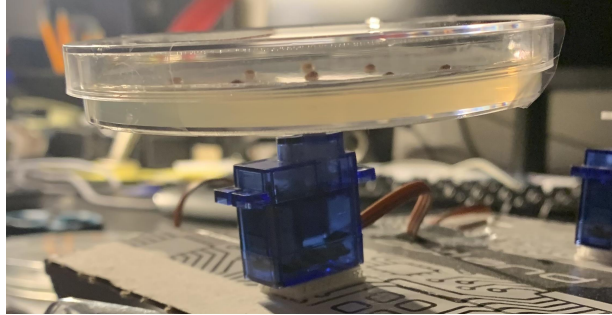


Figure 7. Rotating trial of control group, fully parallel with the ground.

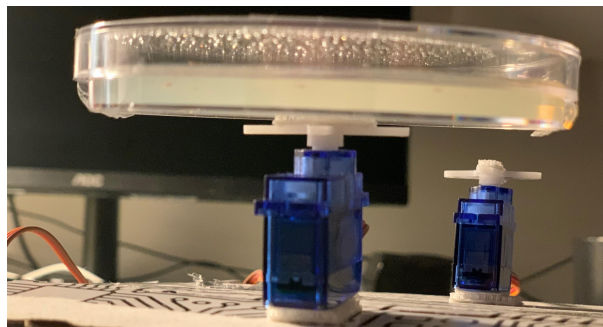


Figure 8. Visible condensation on lid of petri dishes.

BIBLIOGRAPHY

1. Ali, A. S., & Elozeiri, A. A. (2017). Metabolic Processes During Seed Germination. In *www.intechopen.com*. IntechOpen. Retrieved from <https://www.intechopen.com/chapters/56975>
2. Arnaud, C., Bonnot, C., Desnos, T., & Nussaume, L. (2010). The root cap at the forefront. *Comptes Rendus Biologies*, 333(4), 335–343. <https://doi.org/10.1016/j.crvi.2010.01.011>
3. Belyavskaya, N. A. (2004). Biological effects due to weak magnetic field on plants. *Advances in Space Research*, 34(7), 1566–1574. <https://doi.org/10.1016/j.asr.2004.01.021>
4. Brown, A. H., Johnsson, A., Chapman, D., & Heathcote, D. (1996). Gravitropic responses of the Avena coleoptile in space and on clinostats. IV. The clinostat as a substitute for space experiments. *Physiologia Plantarum*, 98(1), 210–214. <https://doi.org/10.1111/j.1399-3054.1996.tb00694.x>
5. Commercial and Marketing Pricing Policy - NASA. (2019, May 31). Retrieved October 2, 2023, from <https://www.nasa.gov/humans-in-space/commercial-and-marketing-pricing-policy/>
6. Dedolph, R. R. (1967). The influence of simulated low-gravity environments on growth, development and metabolism of plants. *Life Sciences and Space*

- Research*, 5, 217–228. Retrieved from <https://pubmed.ncbi.nlm.nih.gov/11973847/>
7. Farooq, M. A., Ma, W., Shen, S., & Gu, A. (2022). Underlying Biochemical and Molecular Mechanisms for Seed Germination. *International Journal of Molecular Sciences*, 23(15), 8502. <https://doi.org/10.3390/ijms23158502>
 8. Hasenstein, K., & Van Loon, J. (n.d.). 4 *Clinostats and Other Rotating Systems-Design, Function, and Limitations*. Retrieved from https://www.riverpublishers.com/pdf/ebook/chapter/RP_9788793237544C14.pdf
 9. Hou, G., Mohamalawari, D. R., & Blancaflor, E. B. (2003). Enhanced Gravitropism of Roots with a Disrupted Cap Actin Cytoskeleton. *Plant Physiology*, 131(3), 1360–1373. <https://doi.org/10.1104/pp.014423>
 10. J. Derek Bewley, & Black, M. (1978). Biochemistry of Germination and Growth. *Springer EBooks*, 132–176. https://doi.org/10.1007/978-3-642-66668-1_5
 11. Kiss, J. Z., Wright, J. B., & Caspar, T. (1996). Gravitropism in roots of intermediate-starch mutants of Arabidopsis. *Physiologia Plantarum*, 97(2), 237–244. <https://doi.org/10.1034/j.1399-3054.1996.970205.x>
 12. Link, B. M., Busse, J. S., & Stankovic, B. (2014). Seed-to-Seed-to-Seed Growth and Development of Arabidopsis in Microgravity. *Astrobiology*, 14(10), 866–875. <https://doi.org/10.1089/ast.2014.1184>
 13. Link, B. M., & Cosgrove, D. J. (1999). Analysis of Peg Formation in Cucumber Seedlings Grown on Clinostats and in a Microgravity (Space) Environment. *Journal of Plant Research*, 112(4), 507–516. <https://doi.org/10.1007/pl00013907>

14. Microgravity : Experiment - International Space Station - JAXA. (n.d.). Retrieved from iss.jaxa.jp website:
<https://iss.jaxa.jp/en/kiboexp/seu/categories/microgravity/index.html>
15. Miyamoto, K., Hoshino, T., Yamashita, M., & Ueda, J. (2005). Automorphosis of etiolated pea seedlings in space is simulated by a three-dimensional clinostat and the application of inhibitors of auxin polar transport. *Physiologia Plantarum*, 123(4), 467–474. <https://doi.org/10.1111/j.1399-3054.2005.00472.x>
16. Muthert, L. W. F., Izzo, L. G., van Zanten, M., & Aronne, G. (2020). Root Tropisms: Investigations on Earth and in Space to Unravel Plant Growth Direction. *Frontiers in Plant Science*, 10. <https://doi.org/10.3389/fpls.2019.01807>
17. Stankovic, B. (2018). Plants in Space. In *www.intechopen.com*. IntechOpen. Retrieved from <https://www.intechopen.com/chapters/59615>
18. Vitha, S., Yang, M., Sack, F. D., & Kiss, J. Z. (2007). Gravitropism in the starch excess mutant of *Arabidopsis thaliana*. *American Journal of Botany*, 94(4), 590–598. <https://doi.org/10.3732/ajb.94.4.590>

List Of Images

- [Figure 1. Dense auxin hormone granules located in columella cells of the root cap \(Kiss J., 1996\)](#)
- [Figure 2. Physiology of the root cap of plants \(Arnaud C., 2010\)](#)
- [Figure 3. The final version of my clinostat containing three motors \(blue boxes\) capable to be set at various angles](#)

- [Figure 4. 3x3 square layout of Lepidium sativum seeds on a Petri dish filled with jelly](#)
- [Graph 1. Averages from table 1 illustrated on a graph with a trend line showcasing the trend.](#)
- [Graph 2. Data from table 1 illustrated on a graph with a polynomial trendline.](#)
- [Graph 3. A trend is drawn on Graph 2 based on the first three groups. An exponential trend line can be clearly observed as a result](#)
- [Figure 5. Rotating trials of the zero-g group and mars group](#)
- [Figure 6. Rotating trials of zero-g group and mars group](#)
- [Figure 7. Rotating trial of control group, fully parallel with the ground](#)
- [Figure 8. Visible condensation on lid of petri dishes](#)

List Of Tables

- [Table 1. Raw data from the experiment containing 9 tests per group of different exposure of g-force along with the averages for each group and the standard deviation. The results were the root's length measured in centimeters](#)